

AI Patent Intelligence: Week 1 report

Dataset: 2248 patent documents from the EPO Open Patent Services (publication years 2010 to 2026), grouped into 1540 patent families.

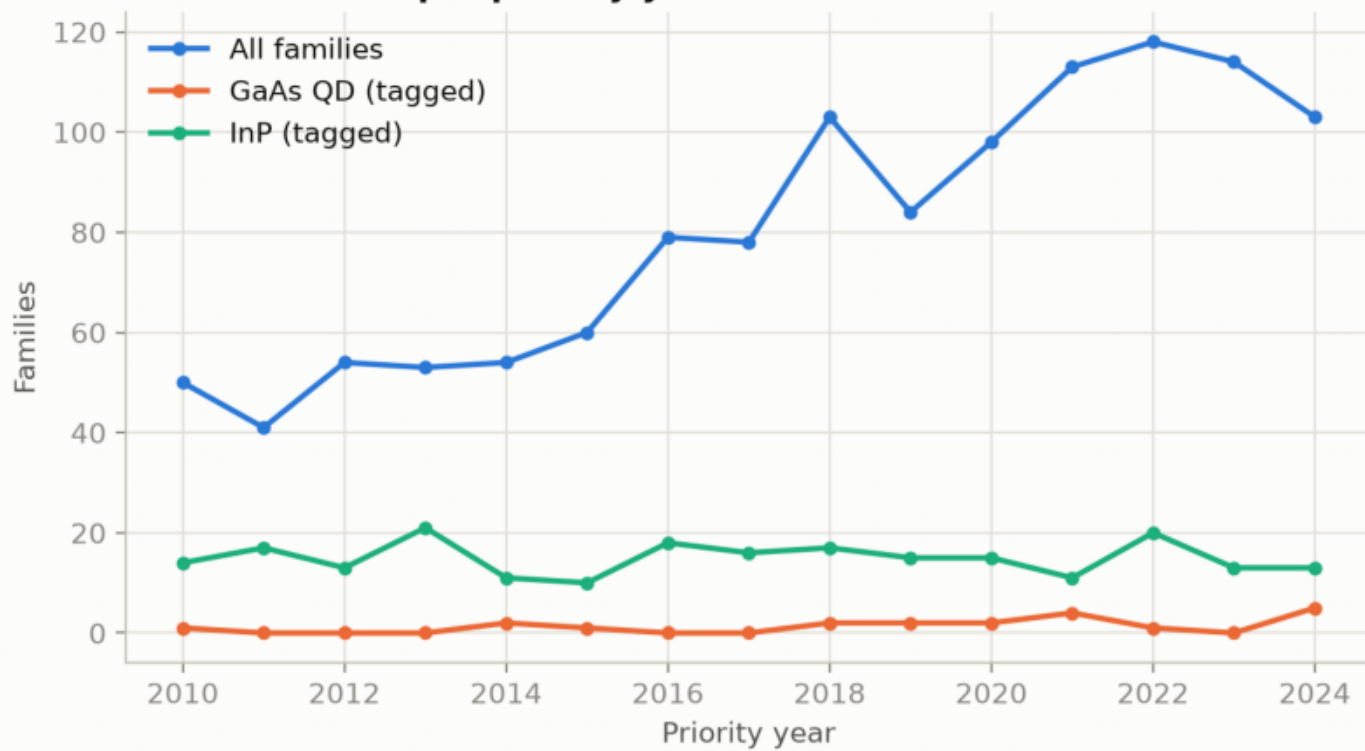
Working set for the charts: 1202 families with priority years 2010 to 2024, without 122 families flagged as GaN or quantum cascade lasers.

Top companies by families (lead applicant): INST SEMICONDUCTORS CAS (51), SUMITOMO ELECTRIC INDUSTRIES (50), NIPPON TELEGRAPH & TELEPHONE (40), MITSUBISHI ELECTRIC (35), ANHUI GEEN SEMICONDUCTOR (32).

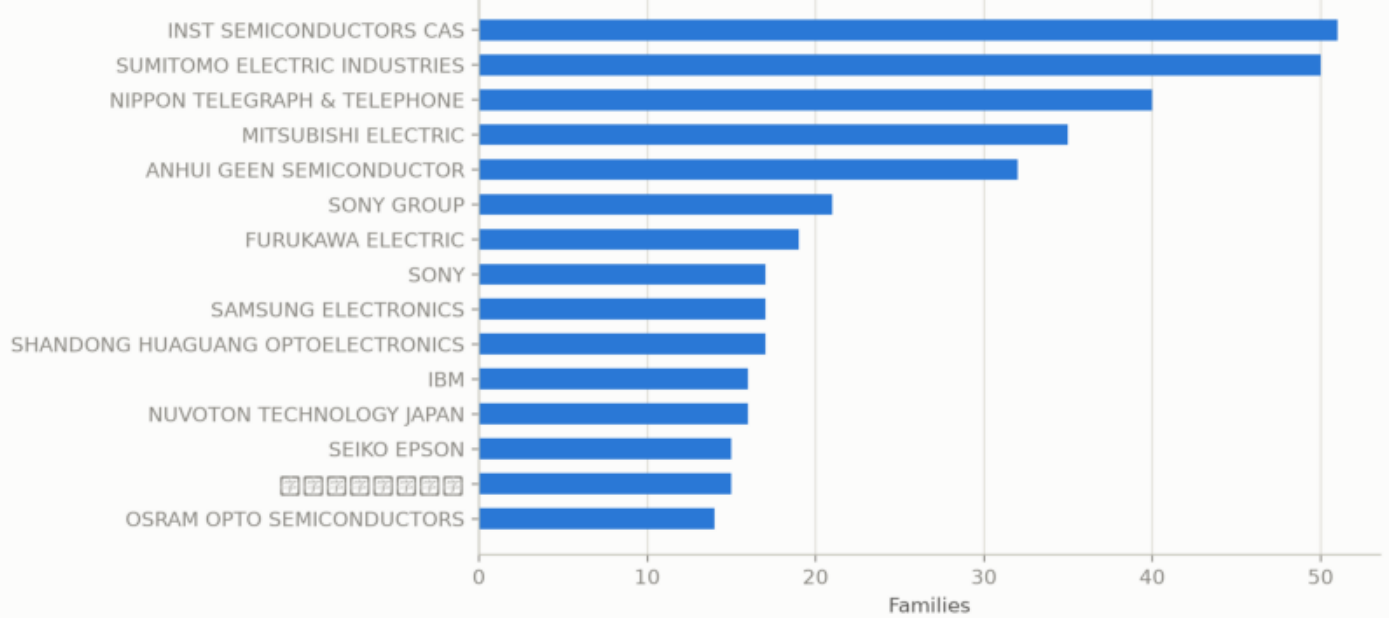
Tag coverage: 20% of families carry a material tag, 16% an emission tag, 5% an application tag.

Limits: tags come from keyword rules on titles, abstracts and CPC codes, not from claims. Years before 2010 are not collected. Company names are only lightly cleaned. Counts follow the EPO's one-result-per-application grouping. The latest priority years are incomplete because patents are published about 18 months after filing.

Patent families per priority year



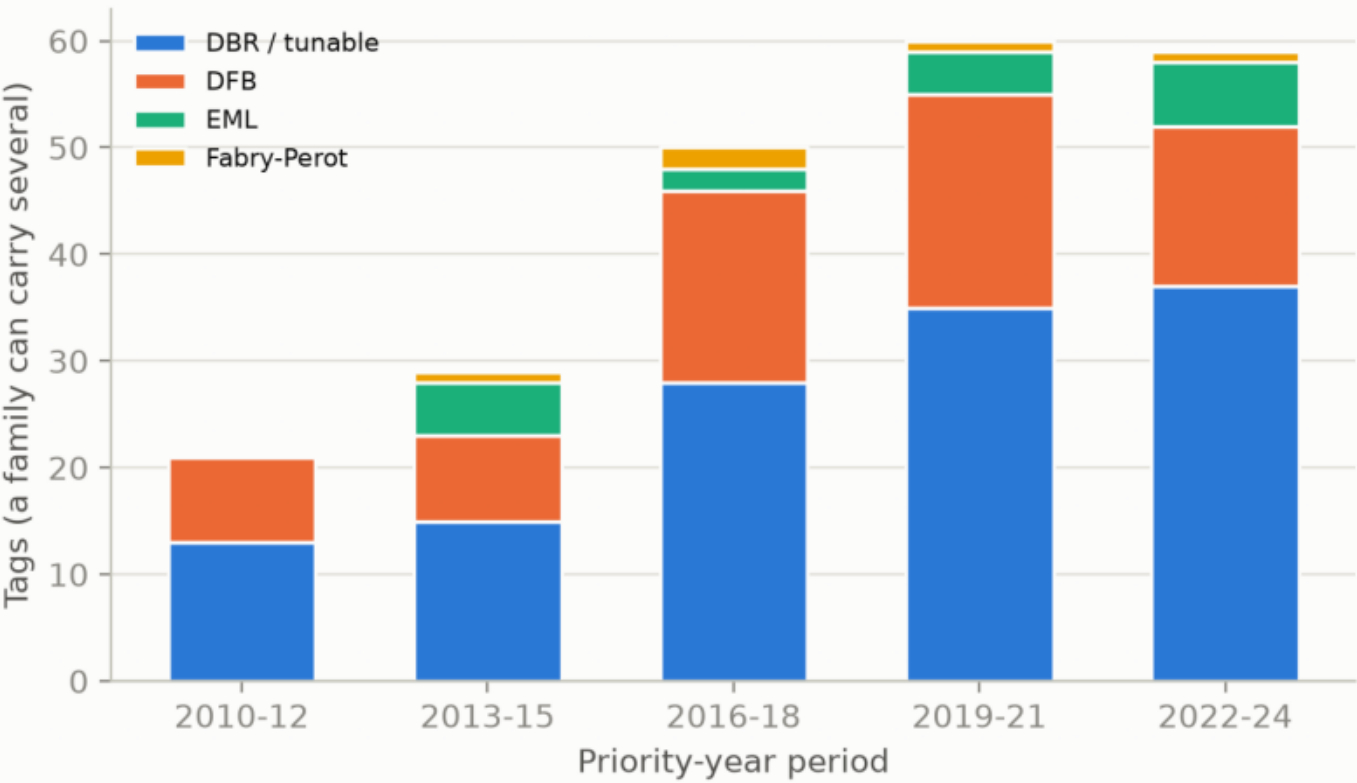
Top 15 companies by patent families (lead applicant)



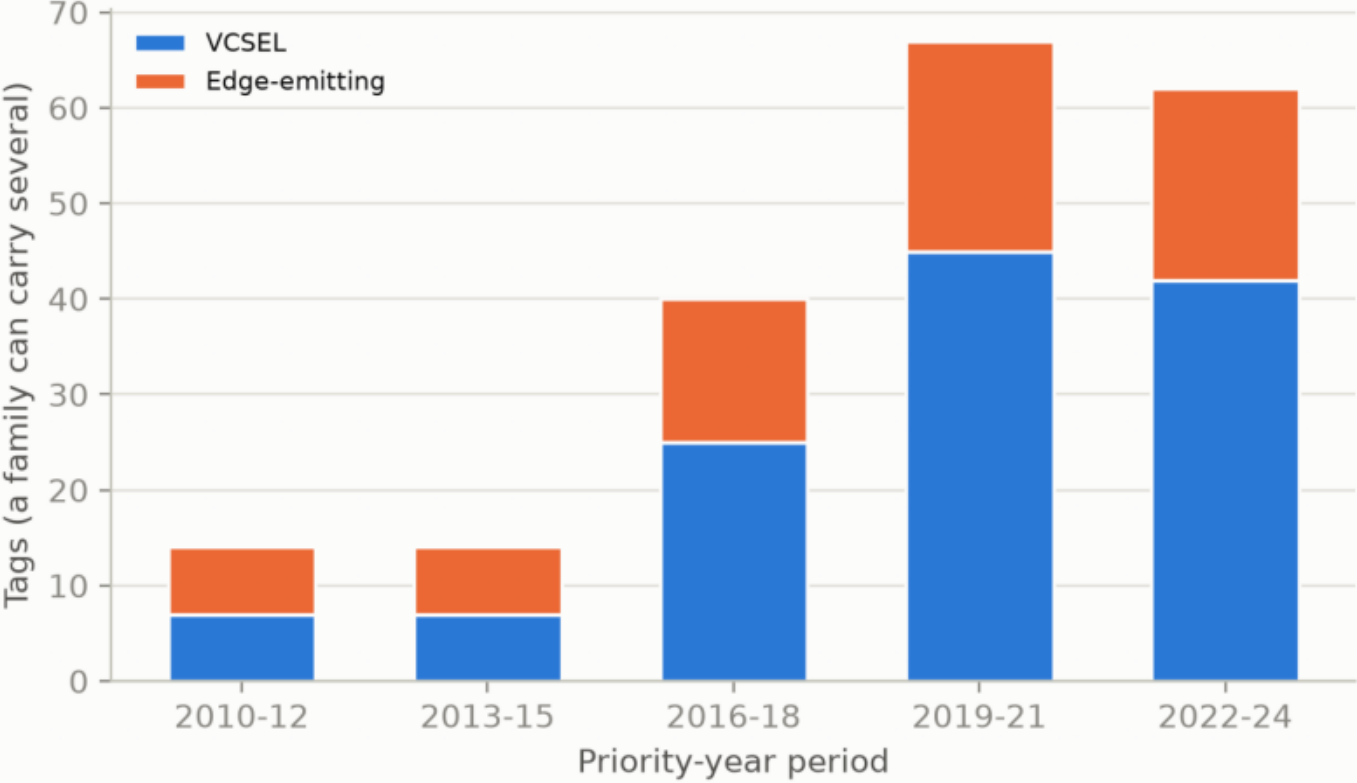
Families per period, top 12 companies

INST SEMICONDUCTORS CAS	1	7	18	13	12
SUMITOMO ELECTRIC INDUSTRIES	14	6	10	12	8
NIPPON TELEGRAPH & TELEPHONE	4	7	4	15	10
MITSUBISHI ELECTRIC	9	12	8	2	4
ANHUI GEEN SEMICONDUCTOR					32
SONY GROUP				10	11
FURUKAWA ELECTRIC	8	2	2	3	4
SONY	1	10	5	1	
SAMSUNG ELECTRONICS		3	7	3	4
SHANDONG HUAGUANG OPTOELECTRONICS		2	7	3	5
IBM		7	7	1	1
NUVOTON TECHNOLOGY JAPAN				2	14
	2010-12	2013-15	2016-18	2019-21	2022-24

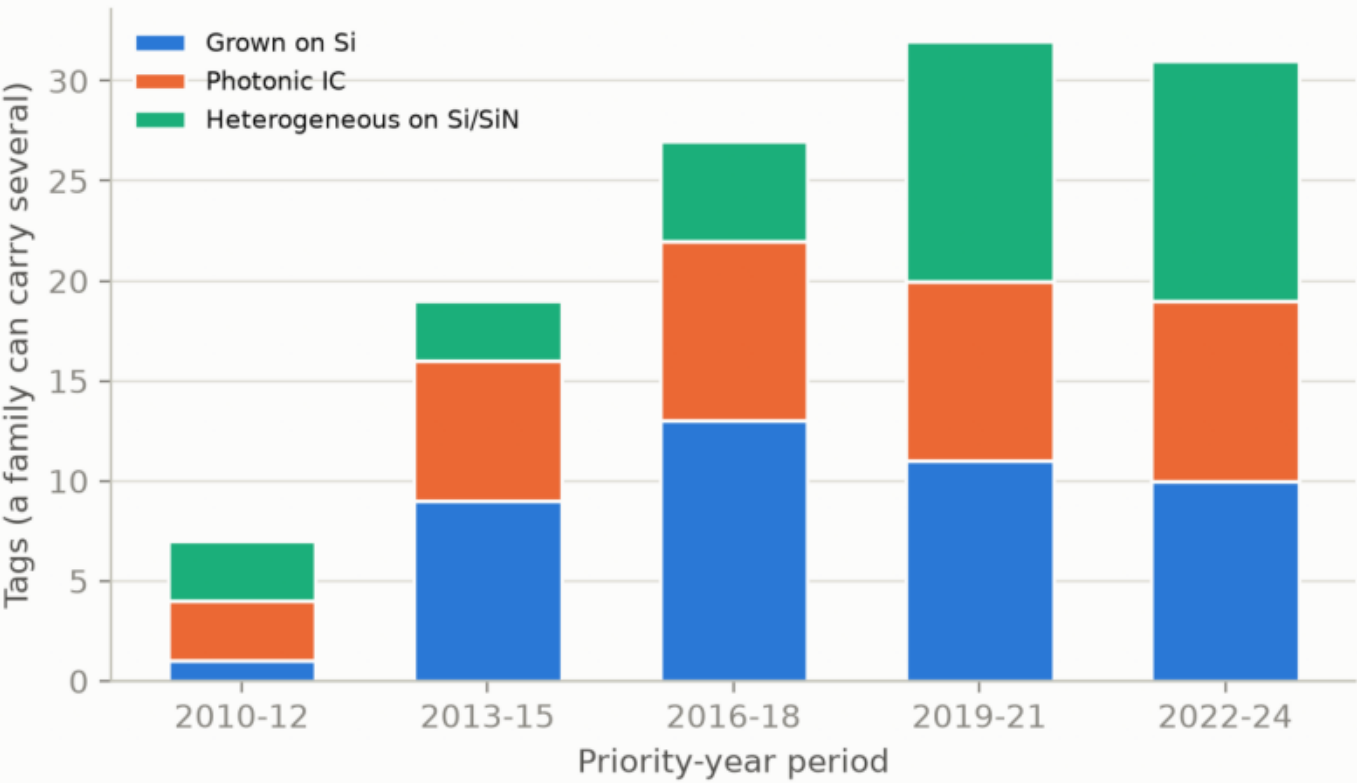
Cavity type over time



Emission type over time



Integration over time



Company x technology tag (families)

INST SEMICONDUCTORS CAS	21	14	5		8	3	2	4	4	5	1	3	1	1
SUMITOMO ELECTRIC INDUSTRIES	7	2	6	4		1				2				
NIPPON TELEGRAPH & TELEPHONE	13	18	4			7	1		4		1	1	2	1
MITSUBISHI ELECTRIC	5	25				1				1				
ANHUI GEEN SEMICONDUCTOR	4													
SONY GROUP		1	3	1			1							
FURUKAWA ELECTRIC	6	1		1		2								
SONY	3													
SAMSUNG ELECTRONICS	2		5	4	1		1	3	2					
SHANDONG HUAGUANG OPTOELECTRONICS	13	3	2	2	1		1						1	1
IBM	3	2						3	1					
NUVOTON TECHNOLOGY JAPAN	2													
	Quantum well	InP	DBR / tunable	VCSEL	Edge-emitting	DFB	Quantum dot	Grown on Si	Photonic IC	Heterogeneous on Si/SiN	Sensing / LIDAR	Telecom	High-temperature	GaAs QD

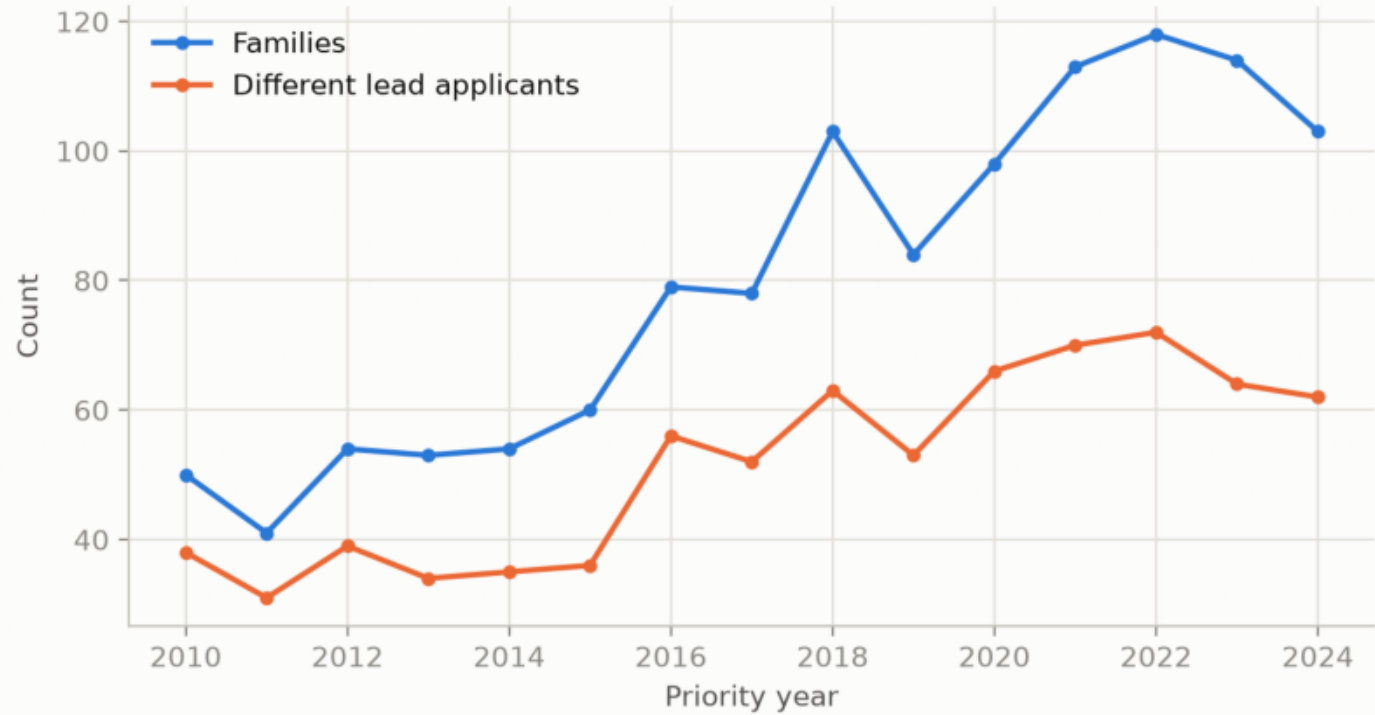
Material x application

InP	0	10	3
GaAs QD	0	2	1
	Datacenter	Telecom	Sensing / LiDAR

Material x integration

InP	7	8	9
GaAs QD	0	0	2
	Photonic IC	Heterogeneous on Si/SiN	Grown on Si

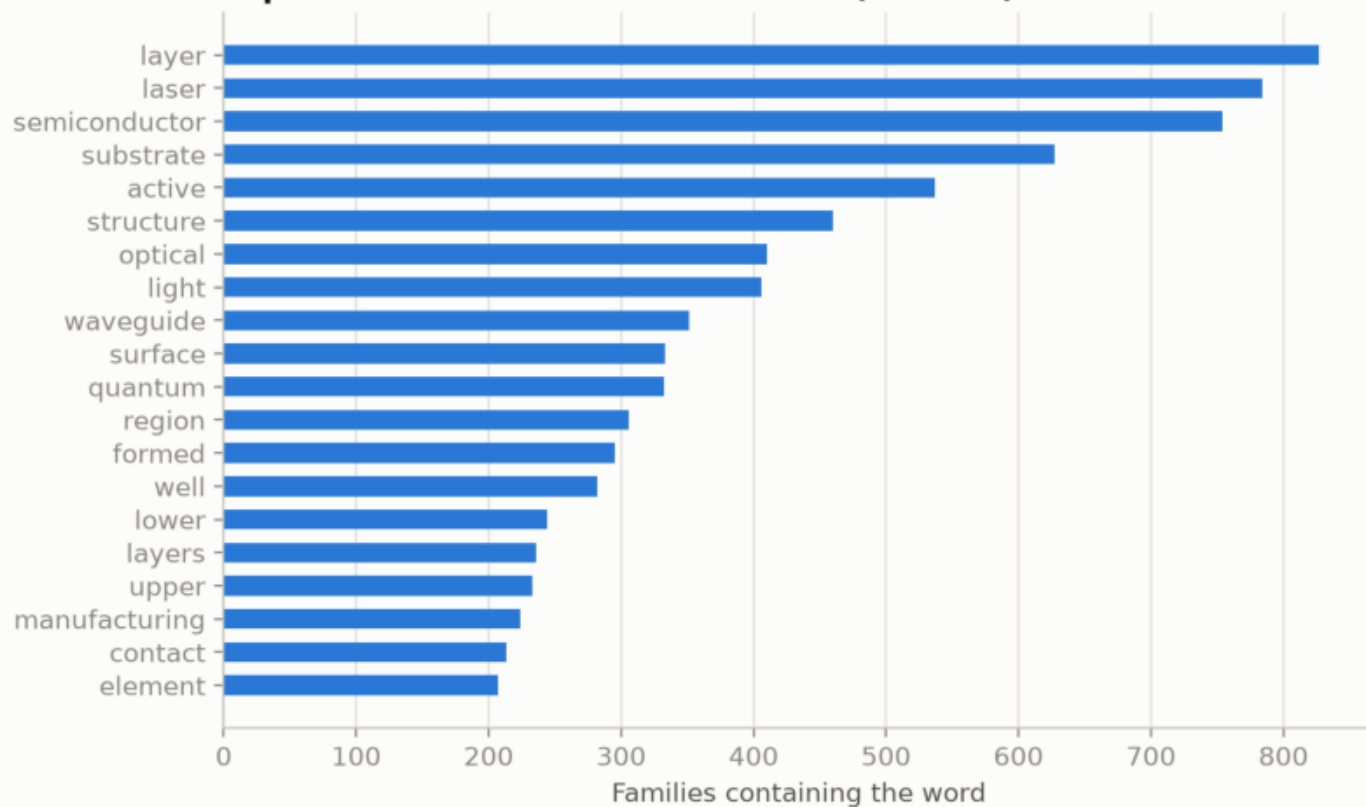
Filings and applicants per priority year



Tracked companies: families per period

Coherent (incl. II-VI, Finisar)	2	2	2	2	3
Lumentum (incl. Oclaro, JDSU)	3	4	13	12	3
ams OSRAM	2	6	3	3	2
Trumpf			1		2
Broadcom (incl. Avago)		2			
Nokia Bell Labs (incl. Alcatel)					2
	2010-12	2013-15	2016-18	2019-21	2022-24

Top 20 words in titles and abstracts (families)



Initial observations

Observation 1: replace this with a finding and its number, for example the growth of families per year.

Observation 2: replace this with a finding about companies.

Observation 3: replace this with a finding about the technology mix.